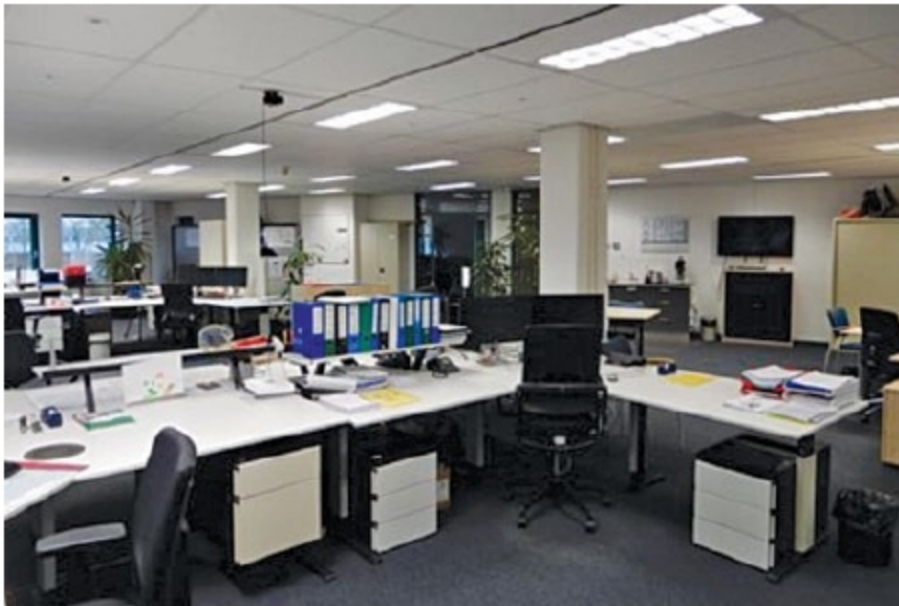


Smart lighting strategy that can save up to 25 per cent energy

Employees often share a large open office with flexible workplaces and working hours, leaving many desks vacant for large parts of the day. But lighting is still on the entire day, in the entire office. Smart lighting with motion detectors can significantly reduce energy consumption in offices, but when light sources are frequently switched on and off, it becomes unpleasant and distracting for the employees.



An example of a Dutch open office without partitions (Credit: Christel de Bakker)

Eindhoven University of Technology researcher Christel de Bakker has developed a method to set lighting in such a way that lighting energy consumption is reduced by more than 25 per cent, without compromising user comfort.

To find out the best lighting strategy for employee comfort, de Bakker set up two darkened research areas of different sizes as an office. She then asked the subjects—95 in total—to set lighting in a way that felt comfortable. Using these results, she then tested the lighting setup that had been most frequently chosen in a real open office, and asked the employees there—20 in total—to evaluate the lighting conditions.

According to de Bakker, test subjects were not able to switch off the light completely in these experiments because earlier research showed that this caused unpleasantly large light contrasts in the area. So, employees could only dim the light.

de Bakker discovered that an office with three light levels is the best strategy. If an office worker is working alone in a medium-sized open-plan office, the desk at which the user is working should be fully lit. Lamps above the three surrounding desks may be slightly dimmed ($35\text{--}65\text{cd/m}^2$), while lighting in the rest of the room may be dimmed even more ($20\text{--}50\text{cd/m}^2$). de Bakker calculated that this would save more than 25 per cent of the energy consumption for lighting in a medium-sized office.

Treating fear of heights using virtual reality

A new study from researchers at Oxford University has used a virtual reality (VR)-based treatment to help people overcome the clinically significant fear of heights. The VR treatment is the first to demonstrate the effectiveness of a completely automated therapy guided by a computer-generated virtual therapist that responds to the patient's voice.

Most prior VR psychological experiments involved a therapist being present to directly guide a patient through a specific experience, but this Oxford study demonstrates an entirely self-contained model with sessions led by a computer-generated therapist.

Subjects in the experimental group received six thirty-

minute VR sessions over a two-week period. The sessions were guided by a virtual coach named Nic, who directed participants through a series of activities designed to affirm the safety of experiences that the subjects perceived as dangerous or anxiety-inducing.

The results were extraordinarily impressive with VR participants showing, on average, a 68 per cent reduction in their fear of heights compared to a randomly assigned control group that received no VR therapy. The automated nature of a treatment method such as this demonstrates a way to potentially roll out widespread therapies that do not need the extra cost and resources demanded by having trained therapists present.

Mitsubishi Heavy Industries develops two firefighting robots

Mitsubishi Heavy Industries has developed two firefighting robots: Water Cannon Robot and Hose Extension Robot. As a team, the duo is expected to play an active role in situations that are extremely hazardous for firefighting crews, such as fires at petrochemical plants. Water Cannon Robot can extinguish and cool fires where human intervention is difficult, while Hose Extension Robot can automatically lay out up to 300m of fire hose to supply water to Water Cannon Robot.

Together, the two robots—when integrated with a reconnaissance and surveillance robot (available in aerial and

ground models) and a command system—constitute a Firefighting Robot System. The system is designed for installation on a dedicated transport vehicle that can be brought directly to the location of the fire.

The two robots are built on chassis frames modified from small farm buggies that offer strong suspension combined with outstanding road manoeuvring. Their frames are equipped with GPS and laser sensors that together enable autonomously controllable movement. With the integration of advanced technologies, the robots are able to self-drive directly to the scene of a fire, equipped with an attached firehose.